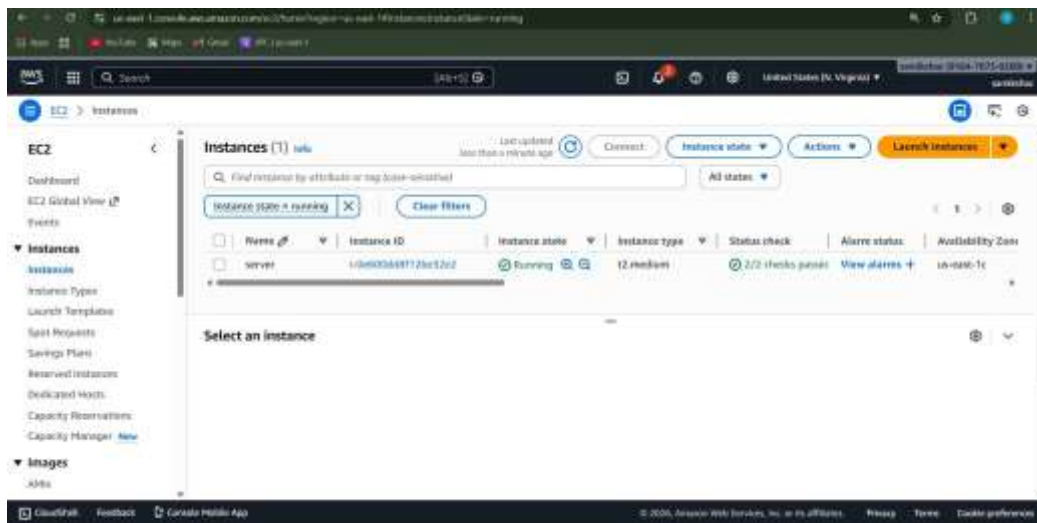


1. Application

- Simple frontend + backend app
- Use one database (MySQL / MongoDB / Influx DB)
- App must connect to the database successfully



```
Windows PowerShell root@ip-172-31-21-55:/home/

httpd-tools x86_64 2.4.66-1.amzn2023.0.1 amazonlinux 81 k
libbrotli x86_64 1.0.9-4.amzn2023.0.2 amazonlinux 315 k
mailcap noarch 2.1.49-3.amzn2023.0.3 amazonlinux 33 k
Installing weak dependencies:
apr-util-openssl x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 15 k
mod_http2 x86_64 2.0.27-1.amzn2023.0.3 amazonlinux 166 k
mod_lua x86_64 2.4.66-1.amzn2023.0.1 amazonlinux 60 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64 2.1 MB/s | 97 kB 00:00
(2/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 536 kB/s | 15 kB 00:00
(3/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 161 kB/s | 13 kB 00:00
(4/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 1.4 MB/s | 129 kB 00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.1.x86_64 703 kB/s | 19 kB 00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64 2.0 MB/s | 47 kB 00:00
(7/13): httpd-filesystem-2.4.66-1.amzn2023.0.1.x86_64 462 kB/s | 13 kB 00:00
(8/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 2.7 MB/s | 81 kB 00:00
(9/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64 22 MB/s | 1.4 MB 00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10 MB/s | 315 kB 00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch 1.1 MB/s | 33 kB 00:00
(12/13): mod_http2-2.0.27-1.amzn2023.0.3.x86_64 6.6 MB/s | 166 kB 00:00
(13/13): mod_lua-2.4.66-1.amzn2023.0.1.x86_64 2.2 MB/s | 60 kB 00:00
-----
Total 10 MB/s | 2.4 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Installing : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 2/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
```

```
Windows PowerShell X root@ip-172-31-21-55:/home/ X + v - □ X
Verifying      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noa 5/13
Verifying      : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13
Verifying      : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13
Verifying      : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Verifying      : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13
Verifying      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying      : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying      : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13
=====
WARNING:
A newer release of "Amazon Linux" is available.

Available Versions:

Version 2023.10.20260202:
Run the following command to upgrade to 2023.10.20260202:

dnf upgrade --releasever=2023.10.20260202

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10
.20260202.html
=====
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64
httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64
```

```
Windows PowerShell  root@ip-172-31-21-55:/home/ + - □ ×

Complete!
[root@ip-172-31-21-55 ec2-user]# sudo systemctl start httpd
[root@ip-172-31-21-55 ec2-user]# sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service →
/usr/lib/systemd/system/httpd.service.
[root@ip-172-31-21-55 ec2-user]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset=
   Active: active (running) since Sat 2026-02-07 03:48:55 UTC; 17s ago
     Docs: man:httpd.service(8)
  Main PID: 30668 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; B
    Tasks: 177 (limit: 4666)
   Memory: 13.0M
      CPU: 73ms
   CGroup: /system.slice/httpd.service
           └─30668 /usr/sbin/httpd -DFOREGROUND
             └─30669 /usr/sbin/httpd -DFOREGROUND
               └─30670 /usr/sbin/httpd -DFOREGROUND
                 └─30671 /usr/sbin/httpd -DFOREGROUND
                   └─30672 /usr/sbin/httpd -DFOREGROUND

Feb 07 03:48:55 ip-172-31-21-55.ec2.internal systemd[1]: Starting httpd.serv
Feb 07 03:48:55 ip-172-31-21-55.ec2.internal httpd[30668]: Server configure
Feb 07 03:48:55 ip-172-31-21-55.ec2.internal systemd[1]: Started httpd.serv
lines 1-19/19 (END)
[root@ip-172-31-21-55 ec2-user]# sudo yum install mariadb-server -y
Last metadata expiration check: 0:34:53 ago on Sat Feb  7 03:17:37 2026.
No match for argument: mariadb-server
Error: Unable to find a match: mariadb-server
[root@ip-172-31-21-55 ec2-user]# cat /etc/os-release
NAME="Amazon Linux"
VERSION="2023"
ID="amzn"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform:al2023"
PRETTY_NAME="Amazon Linux 2023.10.20260120"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2023"
```

```
Windows PowerShell
root@ip-172-31-21-55:/home/

VERSION="2023"
ID="amzn"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform:al2023"
PRETTY_NAME="Amazon Linux 2023.10.20260120"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2023"
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_NAME="AWS"
VENDOR_URL="https://aws.amazon.com/"
SUPPORT_END="2029-06-30"
[root@ip-172-31-21-55 ec2-user]# sudo dnf install mariadb105-server -y
Last metadata expiration check: 0:35:40 ago on Sat Feb  7 03:17:37 2026.
Dependencies resolved.
=====
Package                                Arch    Version                                Repository    Size
=====
Installing:
mariadb105-server                      x86_64  3:10.5.29-1.amzn2023.0.1             amazonlinux   10 M
Installing dependencies:
mariadb-connector-c                    x86_64  3.3.10-1.amzn2023.0.1               amazonlinux   211 k
mariadb-connector-c-config              noarch  3.3.10-1.amzn2023.0.1               amazonlinux    9.9 k
mariadb105                             x86_64  3:10.5.29-1.amzn2023.0.1             amazonlinux   1.5 M
mariadb105-common                      x86_64  3:10.5.29-1.amzn2023.0.1             amazonlinux    28 k
mariadb105-errmsg                      x86_64  3:10.5.29-1.amzn2023.0.1             amazonlinux   212 k
mysql-selinux                          noarch  1.0.4-2.amzn2023.0.3                 amazonlinux    36 k
perl-B                                  x86_64  1.80-477.amzn2023.0.7                amazonlinux   177 k
perl-DBD-MariaDB                      x86_64  1.22-1.amzn2023.0.4                  amazonlinux   153 k
perl-DBI                               x86_64  1.643-7.amzn2023.0.3                  amazonlinux   700 k
perl-Data-Dumper                      x86_64  2.174-460.amzn2023.0.2                amazonlinux    55 k
perl-File-Copy                         noarch  2.34-477.amzn2023.0.7                amazonlinux    20 k
perl-FileHandle                       noarch  2.03-477.amzn2023.0.7                amazonlinux    15 k
perl-Math-BigInt                       noarch  1:1.9998.39-2.amzn2023.0.2            amazonlinux   202 k
perl-Math-BigRat                       noarch  0.2624-500.amzn2023.0.2              amazonlinux    42 k
perl-Math-Complex                      noarch  1.59-477.amzn2023.0.7                amazonlinux    46 k
```



```
Windows PowerShell x root@ip-172-31-21-55/home/ x + - [X]
mariadb105-backup x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 6.0 M
mariadb105-cracklib-password-check x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 13 k
mariadb105-gssapi-server x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 15 k
mariadb105-server-utils x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 207 k

Transaction Summary
=====
Install 22 Packages

Total download size: 20 M
Installed size: 117 M
Downloading Packages:
(1/22): mariadb-connector-c-3.3.10-1.amzn20 5.0 MB/s | 211 kB 00:00
(2/22): mariadb-connector-c-config-3.3.10-1 225 kB/s | 9.9 kB 00:00
(3/22): mariadb105-10.5.29-1.amzn2023.0.1.x 24 MB/s | 1.5 MB 00:00
(4/22): mariadb105-common-10.5.29-1.amzn202 979 kB/s | 28 kB 00:00
(5/22): mariadb105-cracklib-password-check- 565 kB/s | 13 kB 00:00
(6/22): mariadb105-errmsg-10.5.29-1.amzn202 5.7 MB/s | 212 kB 00:00
(7/22): mariadb105-gssapi-server-10.5.29-1. 513 kB/s | 15 kB 00:00
(8/22): mariadb105-backup-10.5.29-1.amzn202 55 MB/s | 6.8 MB 00:00
(9/22): mariadb105-server-utils-10.5.29-1.a 5.2 MB/s | 207 kB 00:00
(10/22): mysql-selinux-1.0.4-2.amzn2023.0.3 1.3 MB/s | 36 kB 00:00
(11/22): perl-B-1.80-477.amzn2023.0.7.x86_6 5.1 MB/s | 177 kB 00:00
(12/22): perl-DBD-MariaDB-1.22-1.amzn2023.0 6.1 MB/s | 153 kB 00:00
(13/22): perl-Data-Dumper-2.174-460.amzn202 2.0 MB/s | 55 kB 00:00
(14/22): perl-DBI-1.643-7.amzn2023.0.3.x86_ 13 MB/s | 700 kB 00:00
(15/22): perl-File-Copy-2.34-477.amzn2023.0 794 kB/s | 28 kB 00:00
(16/22): mariadb105-server-10.5.29-1.amzn20 52 MB/s | 10 MB 00:00
(17/22): perl-FileHandle-2.03-477.amzn2023. 234 kB/s | 15 kB 00:00
(18/22): perl-Math-BigInt-1.9998.39-2.amzn2 3.4 MB/s | 202 kB 00:00
(19/22): perl-Math-BigRat-0.2624-500.amzn20 1.5 MB/s | 42 kB 00:00
(20/22): perl-Math-Complex-1.59-477.amzn202 1.6 MB/s | 46 kB 00:00
(21/22): perl-Sys-Hostname-1.23-477.amzn202 627 kB/s | 16 kB 00:00
(22/22): perl-base-2.27-477.amzn2023.0.7.no 630 kB/s | 16 kB 00:00
-----
Total 49 MB/s | 20 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
```

```
Windows PowerShell
root@ip-172-31-21-53:/home/

Verifying      : perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64      13/22
Verifying      : perl-DBI-1.643-7.amzn2023.0.3.x86_64           14/22
Verifying      : perl-Data-Dumper-2.174-460.amzn2023.0.2.x86_64  15/22
Verifying      : perl-File-Copy-2.34-477.amzn2023.0.7.noarch     16/22
Verifying      : perl-FileHandle-2.03-477.amzn2023.0.7.noarch    17/22
Verifying      : perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.no  18/22
Verifying      : perl-Math-BigRat-0.2624-580.amzn2023.0.2.noarc  19/22
Verifying      : perl-Math-Complex-1.59-477.amzn2023.0.7.noarch 20/22
Verifying      : perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64 21/22
Verifying      : perl-base-2.27-477.amzn2023.0.7.noarch         22/22
=====
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.10.20260202:
  Run the following command to upgrade to 2023.10.20260202:

      dnf upgrade --releasever=2023.10.20260202

  Release notes:
  https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10
  .20260202.html
=====
Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64
  mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
  mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-backup-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-cracklib-password-check-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x86_64
  mysql-selinux-1.0.4-2.amzn2023.0.3.noarch
  perl-B-1.89-477.amzn2023.0.7.x86_64
  perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64
```

```
Windows PowerShell X root@ip-172-31-21-55/home/ X + - [ ] X

perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64
perl-base-2.27-477.amzn2023.0.7.noarch

Complete!
[root@ip-172-31-21-55 ec2-user]# sudo systemctl start mariadb
[root@ip-172-31-21-55 ec2-user]# sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/
mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system
/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service
→ /usr/lib/systemd/system/mariadb.service.
[root@ip-172-31-21-55 ec2-user]# sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] y
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
```

```
Windows PowerShell x root@ip-172-31-21-55:/home/ x + v - □ x

... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
[root@ip-172-31-21-55 ec2-user]# sudo amazon-linux-extras enable php8.0
sudo: amazon-linux-extras: command not found
[root@ip-172-31-21-55 ec2-user]# amazon-linux-extras enable php8.0
bash: amazon-linux-extras: command not found
[root@ip-172-31-21-55 ec2-user]# sudo amazon-linux-extras enable php8.0
sudo: amazon-linux-extras: command not found
[root@ip-172-31-21-55 ec2-user]# sudo dnf update -y
Last metadata expiration check: 0:38:27 ago on Sat Feb  7 03:17:37 2026.
=====
WARNING:
  A newer release of "Amazon Linux" is available.
```



```
Windows PowerShell X root@ip-172-31-21-55/home/ X + - □ X
[root@ip-172-31-21-55 ec2-user]# sudo dnf update -y
Last metadata expiration check: 0:38:27 ago on Sat Feb  7 03:17:37 2026.
=====
WARNING:
  A newer release of "Amazon Linux" is available.

Available Versions:

Version 2023.10.20260202:
  Run the following command to upgrade to 2023.10.20260202:

      dnf upgrade --releasever=2023.10.20260202

Release notes:
  https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10
  .20260202.html
=====
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-21-55 ec2-user]# sudo dnf install httpd -y
Last metadata expiration check: 0:38:32 ago on Sat Feb  7 03:17:37 2026.
Package httpd-2.4.66-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-21-55 ec2-user]# sudo systemctl start httpd
sudo systemctl enable httpd
[root@ip-172-31-21-55 ec2-user]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset=
   Active: active (running) since Sat 2026-02-07 03:48:55 UTC; 7min ago
     Docs: man:httpd.service(8)
   Main PID: 30668 (httpd)
    Status: "Total requests: 3; Idle/Busy workers 100/0; Requests/sec: 0.00
      Tasks: 230 (limit: 4666)
     Memory: 16.1M
        CPU: 601ms
    CGroup: /system.slice/httpd.service
            └─30668 /usr/sbin/httpd -DFOREGROUND
```

```
Windows PowerShell
root@ip-172-31-21-55:/home/

Memory: 16.1M
CPU: 601ms
CGroup: /system.slice/httpd.service
├─30668 /usr/sbin/httpd -DFOREGROUND
├─30669 /usr/sbin/httpd -DFOREGROUND
├─30670 /usr/sbin/httpd -DFOREGROUND
├─30671 /usr/sbin/httpd -DFOREGROUND
├─30672 /usr/sbin/httpd -DFOREGROUND
└─30931 /usr/sbin/httpd -DFOREGROUND

Feb 07 03:48:55 ip-172-31-21-55.ec2.internal systemd[1]: Starting httpd.serv
Feb 07 03:48:55 ip-172-31-21-55.ec2.internal httpd[30668]: Server configure>
Feb 07 03:48:55 ip-172-31-21-55.ec2.internal systemd[1]: Started httpd.serv>

[root@ip-172-31-21-55 ec2-user]# sudo dnf search php
Last metadata expiration check: 0:39:47 ago on Sat Feb  7 03:17:37 2026.
===== Name & Summary Matched: php =====
php-pear.noarch : PHP Extension and Application Repository framework
php8.1.x86_64 : PHP scripting language for creating dynamic web sites
php8.1-bcmath.x86_64 : A module for PHP 8.1 applications for using the
                    : bcmath library
php8.1-cli.x86_64 : Command-line interface for PHP 8.1
php8.1-common.x86_64 : Common files for PHP 8.1
php8.1-dba.x86_64 : A database abstraction layer module for PHP 8.1
                    : applications
php8.1-dbg.x86_64 : The interactive PHP 8.1 debugger
php8.1-devel.x86_64 : Files needed for building PHP 8.1 extensions
php8.1-embedded.x86_64 : PHP 8.1 library for embedding in applications
php8.1-enchant.x86_64 : Enchant spelling extension for PHP 8.1 applications
php8.1-fpm.x86_64 : PHP 8.1 FastCGI Process Manager
php8.1-gd.x86_64 : A module for PHP 8.1 applications for using the gd
                    : graphics library
php8.1-gmp.x86_64 : A module for PHP 8.1 applications for using the GNU MP
                    : library
php8.1-intl.x86_64 : Internationalization extension for PHP 8.1 applications
php8.1-ldap.x86_64 : A module for PHP 8.1 applications that use LDAP
php8.1-mbstring.x86_64 : A module for PHP 8.1 applications which need
                    : multi-byte string handling
php8.1-mysqlnd.x86_64 : A module for PHP 8.1 applications that use MySQL
                    : databases
php8.1-odbc.x86_64 : A module for PHP 8.1 applications that use ODBC
```

```
Windows PowerShell  X  root@ip-172-31-21-55:/home/  X  +  -  □  X
php8.3-embedded.x86_64 : PHP 8.3 library for embedding in applications
php8.3-enchanted.x86_64 : Enchant spelling extension for PHP 8.3 applications
php8.3-fpm.x86_64 : PHP 8.3 FastCGI Process Manager
php8.3-gd.x86_64 : A module for PHP 8.3 applications for using the gd
                  : graphics library
php8.3-gmp.x86_64 : A module for PHP 8.3 applications for using the GNU MP
                  : library
php8.3-intl.x86_64 : Internationalization extension for PHP 8.3 applications
php8.3-ldap.x86_64 : A module for PHP 8.3 applications that use LDAP
php8.3-mbstring.x86_64 : A module for PHP 8.3 applications which need
                  : multi-byte string handling
php8.3-modphp.x86_64 : PHP 8.3 module for Apache HTTP Server
php8.3-mysqlnd.x86_64 : A module for PHP 8.3 applications that use MySQL
                  : databases
php8.3-odbc.x86_64 : A module for PHP 8.3 applications that use ODBC
                  : databases
php8.3-pdo.x86_64 : A database access abstraction module for PHP 8.3
                  : applications
php8.3-pecl-igbinary.x86_64 : Replacement for the standard PHP serializer
php8.3-pecl-redis6.x86_64 : PHP extension for interfacing with key-value
                  : stores
php8.3-pgsql.x86_64 : A PostgreSQL database module for PHP 8.3
php8.3-process.x86_64 : Modules for PHP 8.3 script using system process
                  : interfaces
php8.3-pspell.x86_64 : A module for PHP 8.3 applications for using pspell
                  : interfaces
php8.3-snmp.x86_64 : A module for PHP 8.3 applications that query
                  : SNMP-managed devices
php8.3-soap.x86_64 : A module for PHP 8.3 applications that use the SOAP
                  : protocol
php8.3-tidy.x86_64 : Standard PHP 8.3 module provides tidy library support
php8.3-xml.x86_64 : A module for PHP 8.3 applications which use XML
php8.4.x86_64 : PHP scripting language for creating dynamic web sites
php8.4-bcmath.x86_64 : A module for PHP 8.4 applications for using the
                  : bcmath library
php8.4-cli.x86_64 : Command-line interface for PHP 8.4
php8.4-common.x86_64 : Common files for PHP 8.4
php8.4-dba.x86_64 : A database abstraction layer module for PHP 8.4
                  : applications
php8.4-dbg.x86_64 : The interactive PHP 8.4 debugger
php8.4-devel.x86_64 : Files needed for building PHP 8.4 extensions
```

```
Windows PowerShell
root@ip-172-31-21-55:/home/

php8.4-pecl-igbinary-devel.x86_64 : Igbinary developer files (header)
php8.4-pecl-msgpack.x86_64 : API for communicating with MessagePack
                             : serialization
php8.4-pecl-msgpack-devel.x86_64 : MessagePack developer files (header)
php8.4-sodium.x86_64 : Wrapper for the Sodium cryptographic library
php8.4-zip.x86_64 : ZIP archive management extension
php8.5-ffi.x86_64 : Foreign Function Interface
php8.5-sodium.x86_64 : Wrapper for the Sodium cryptographic library
php8.5-zip.x86_64 : ZIP archive management extension
texlive-graphpaper.noarch : A LaTeX class to generate several types of graph
                             : papers
===== Summary Matched: php =====
composer.noarch : Dependency Manager for PHP
[root@ip-172-31-21-55 ec2-user]# sudo dnf install php php-mysqld -y
Last metadata expiration check: 0:39:55 ago on Sat Feb  7 03:17:37 2026.
Dependencies resolved.
=====
Package           Arch      Version              Repository           Size
=====
Installing:
php8.5            x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          15 k
php8.5-mysqld     x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          155 k
Installing dependencies:
libsodium         x86_64    1.0.19-4.amzn2023   amazonlinux          176 k
libxslt           x86_64    1.1.43-1.amzn2023.0.3 amazonlinux          183 k
nginxfilesystem  noarch    1:1.28.1-1.amzn2023.0.1 amazonlinux          9.6 k
php8.5-cli        x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          6.0 M
php8.5-common     x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          853 k
php8.5-pdo        x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          101 k
php8.5-process    x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          50 k
php8.5-xml        x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          248 k
Installing weak dependencies:
php8.5-fpm        x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          3.0 M
php8.5-openssl    x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          540 k
php8.5-sodium     x86_64    8.5.1-1.amzn2023.0.1 amazonlinux          46 k

Transaction Summary
=====
Install 13 Packages

Total download size: 11 M
```



```
Windows PowerShell
root@ip-172-31-21-55:/home/

Verifying      : php8.5-cli-8.5.1-1.amzn2023.0.1.x86_64      5/13
Verifying      : php8.5-common-8.5.1-1.amzn2023.0.1.x86_64  6/13
Verifying      : php8.5-fpm-8.5.1-1.amzn2023.0.1.x86_64    7/13
Verifying      : php8.5-mbstring-8.5.1-1.amzn2023.0.1.x86_64 8/13
Verifying      : php8.5-mysqlnd-8.5.1-1.amzn2023.0.1.x86_64 9/13
Verifying      : php8.5-pdo-8.5.1-1.amzn2023.0.1.x86_64    10/13
Verifying      : php8.5-process-8.5.1-1.amzn2023.0.1.x86_64 11/13
Verifying      : php8.5-sodium-8.5.1-1.amzn2023.0.1.x86_64  12/13
Verifying      : php8.5-xsl-8.5.1-1.amzn2023.0.1.x86_64     13/13
=====
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.10.20260202:
    Run the following command to upgrade to 2023.10.20260202:

      dnf upgrade --releasever=2023.10.20260202

  Release notes:
    https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10.20260202.html
=====

Installed:
  libsodium-1.0.19-4.amzn2023.x86_64
  libxslt-1.1.43-1.amzn2023.0.3.x86_64
  nginxfilesystem-1:1.28.1-1.amzn2023.0.1.noarch
  php8.5-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-cli-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-common-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-fpm-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-mbstring-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-mysqlnd-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-pdo-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-process-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-sodium-8.5.1-1.amzn2023.0.1.x86_64
  php8.5-xsl-8.5.1-1.amzn2023.0.1.x86_64
```

```
Windows PowerShell X root@ip-172-31-21-55/home/ X + v [ ] X

[root@ip-172-31-21-55 ec2-user]# sudo vi /var/www/html/index.php
"/var/www/html/index.php" [New] 4L, 21B written
[root@ip-172-31-21-55 ec2-user]# sudo systemctl restart httpd
[root@ip-172-31-21-55 ec2-user]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 13
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE lampdb;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]>
MariaDB [(none)]> CREATE USER 'lampuser'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]>
MariaDB [(none)]> GRANT ALL PRIVILEGES ON lampdb.* TO 'lampuser'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]>
MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]>
MariaDB [(none)]> EXIT;
Bye
[root@ip-172-31-21-55 ec2-user]# sudo vi /var/www/html/db.php
[root@ip-172-31-21-55 ec2-user]# sudo systemctl restart httpd
[root@ip-172-31-21-55 ec2-user]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 15
Server version: 10.5.29-MariaDB MariaDB Server
```

```
Windows PowerShell
root@ip-172-31-21-55:/home/

t.

MariaDB [(none)]> CREATE TABLE users (
  ->   id INT AUTO_INCREMENT PRIMARY KEY,
  ->   name VARCHAR(100),
  ->   email VARCHAR(100)
  -> );
ERROR 1046 (3D000): No database selected
MariaDB [(none)]> exit
Bye
[root@ip-172-31-21-55 ec2-user]# sudo vi /var/www/html/config.php
<r/www/html/config.php" [New] 13L, 263B written
[root@ip-172-31-21-55 ec2-user]# sudo vi /var/www/html/insert.php
[root@ip-172-31-21-55 ec2-user]# sudo vi /var/www/html/form.html
<ar/www/html/form.html" [New] 18L, 325B written
[root@ip-172-31-21-55 ec2-user]# sudo systemctl restart httpd
[root@ip-172-31-21-55 ec2-user]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 18
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

t.

MariaDB [(none)]> USE lampdb;
Database changed
MariaDB [lampdb]> SELECT * FROM users;
ERROR 1146 (42S02): Table 'lampdb.users' doesn't exist
MariaDB [lampdb]>
MariaDB [lampdb]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| lampdb |
| mysql |
| performance_schema |
+-----+
```

```
Windows PowerShell x root@ip-172-31-21-55:/home/ x + v
MariaDB [lampdb]>
MariaDB [lampdb]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| lampdb |
| mysql |
| performance_schema |
+-----+
4 rows in set (0.000 sec)

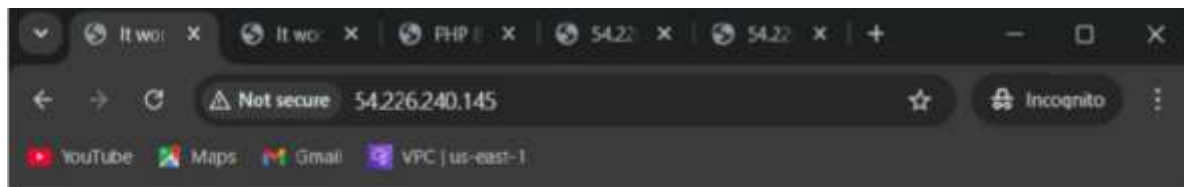
MariaDB [lampdb]> USE lampdb;
Database changed
MariaDB [lampdb]> SHOW TABLES;
Empty set (0.000 sec)

MariaDB [lampdb]> CREATE TABLE users (
-> id INT AUTO_INCREMENT PRIMARY KEY,
-> name VARCHAR(100),
-> email VARCHAR(100)
-> );
Query OK, 0 rows affected (0.011 sec)

MariaDB [lampdb]> SHOW TABLES;
+-----+
| Tables_in_lampdb |
+-----+
| users |
+-----+
1 row in set (0.001 sec)

MariaDB [lampdb]> SELECT * FROM users;
+----+-----+-----+
| id | name | email |
+----+-----+-----+
| 1 | Samiksha Choudhari | choudharisamiksha276@gmail.com |
+----+-----+-----+
1 row in set (0.000 sec)

MariaDB [lampdb]>
```

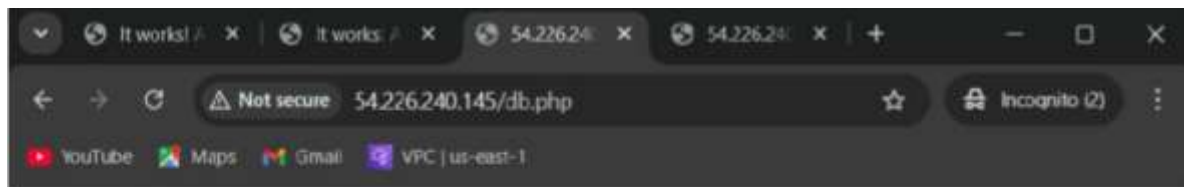



It works!

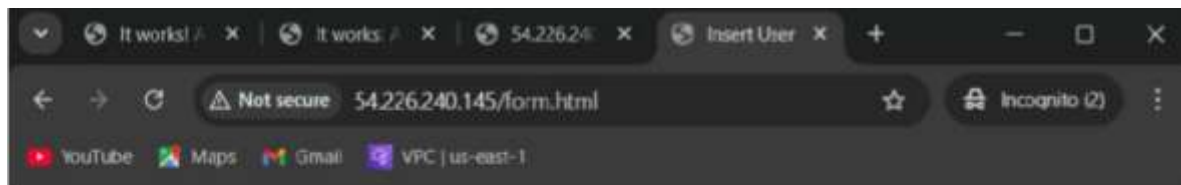


The screenshot shows a web browser window with the address bar displaying 'http://localhost:8080/phpinfo.php'. The page title is 'PHP Version 8.5.1'. The page content is a table listing various system and PHP details.

PHP Version 8.5.1	
System	Linux 5.15.0-101-generic x86_64 GNU/Linux
Build Date	Dec 18 2024 15:24:07
Build System	Linux
Build Provider	Debian
Configure	./configure --with-freetype --with-gd --with-gettext --with-gmp --with-icu --with-ldap --with-libpng --with-libxml --with-mcrypt --with-mhash --with-mysql --with-mysqli --with-openssl --with-pdo-mysql --with-pdo-sqlite --with-sqlite3 --with-xmlrpc --with-zlib
Architecture	x86_64
Server API	Apache2
Virtual Directory Support	Enabled
Configuration File(s)	/etc/php/8.5/apache2/php.ini
Loaded Configuration File	/etc/php/8.5/apache2/php.ini
Scan this dir for additional ini files	/usr/lib/php/8.5
Additional ini files	None
PHP API	20240924
PHP Extension	Zend Engine
Zend Extension	Zend Engine
Zend Extension Build	3.0.0
PHP Extension Build	3.0.0
PHP Major Version	8.5
PHP Minor Version	5
PHP Patch Version	1
PHP Build System	Linux
PHP Build Provider	Debian
PHP Build System	Linux
PHP Build Provider	Debian



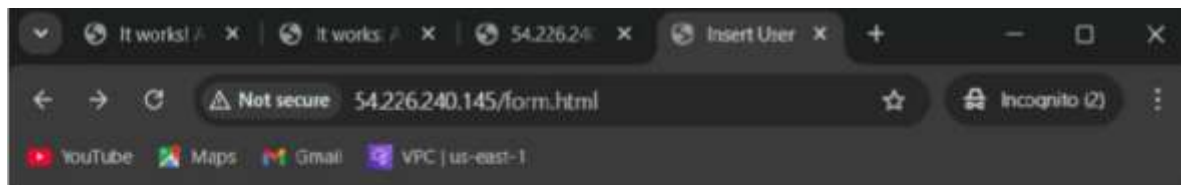
Database Connected Successfully



Add User

Name:

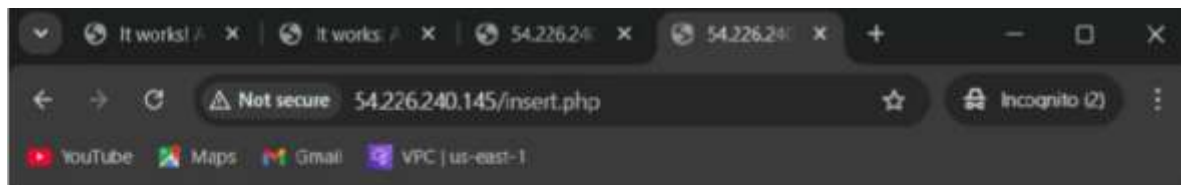
Email:



Add User

Name:

Email:



Record inserted successfully

```

MariaDB [(none)]> SHOW TABLES;
ERROR 1046 (3D0000): No database selected
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| lampdb |
| mysql |
| performance_schema |
+-----+
4 rows in set (0.000 sec)

MariaDB [(none)]> use lampdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [lampdb]> SELECT * FROM users;
+----+-----+-----+
| id | name          | email |
+----+-----+-----+
| 1  | Samiksha Choudhari | choudharisaniksha276@gmail.com |
| 2  | ram            | ran@gmail.com |
| 3  | ram            | ran@gmail.com |
+----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [lampdb]> |

```

Vi index.php



```

ec2-user@ip-172-31-29-70:~
<?php
phpinfo();
?>

```

Vi db.php

```
ec2-user@ip-172-31-29-70:~  
<?php  
$conn = new mysqli("localhost", "lampuser", "password123", "lampdb");  
  
if ($conn->connect_error) {  
    die("Connection failed: " . $conn->connect_error);  
}  
  
echo "Database Connected Successfully";  
?>
```

Vi config.php

```
ec2-user@ip-172-31-29-70:~  
<?php  
$servername = "localhost";  
$username = "lampuser";  
$password = "password123";  
$database = "lampdb";  
  
$conn = new mysqli($servername, $username, $password, $database);  
  
if ($conn->connect_error) {  
    die("Connection failed: " . $conn->connect_error);  
}  
?>
```

Vi insert.php

```
ec2-user@ip-172-31-29-70:~  
<?php  
include 'config.php';  
  
if ($_SERVER["REQUEST_METHOD"] == "POST") {  
    $name = $_POST['name'];  
    $email = $_POST['email'];  
  
    $sql = "INSERT INTO users (name, email) VALUES ('$name', '$email')";  
  
    if ($conn->query($sql) === TRUE) {  
        echo "Record inserted successfully";  
    } else {  
        echo "Error: " . $conn->error;  
    }  
}  
  
$conn->close();  
?>
```

Vi form.php

```
ec2-user@ip-172-31-29-70:~  
<!--  
<!DOCTYPE html>  
<html>  
<head>  
    <title>Insert User</title>  
</head>  
<body>  
  
<h2>Add User</h2>  
  
<form action="insert.php" method="POST">  
    Name: <input type="text" name="name" required><br><br>  
    Email: <input type="email" name="email" required><br><br>  
    <input type="submit" value="Submit">  
</form>  
  
</body>  
</html>  
|  
~  
~
```